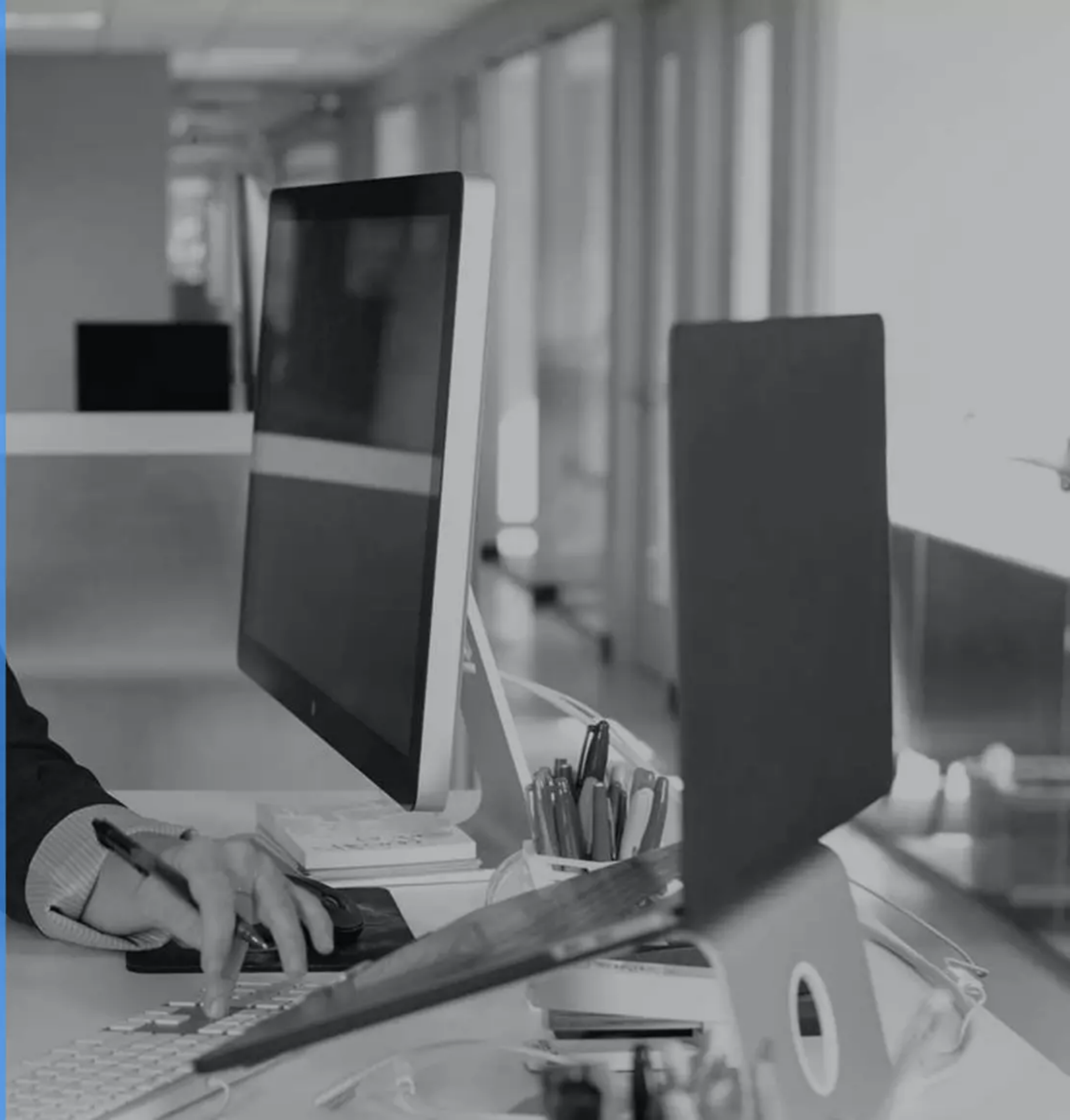




Introduction to Apache Kafka

By: Dhaval Thumar

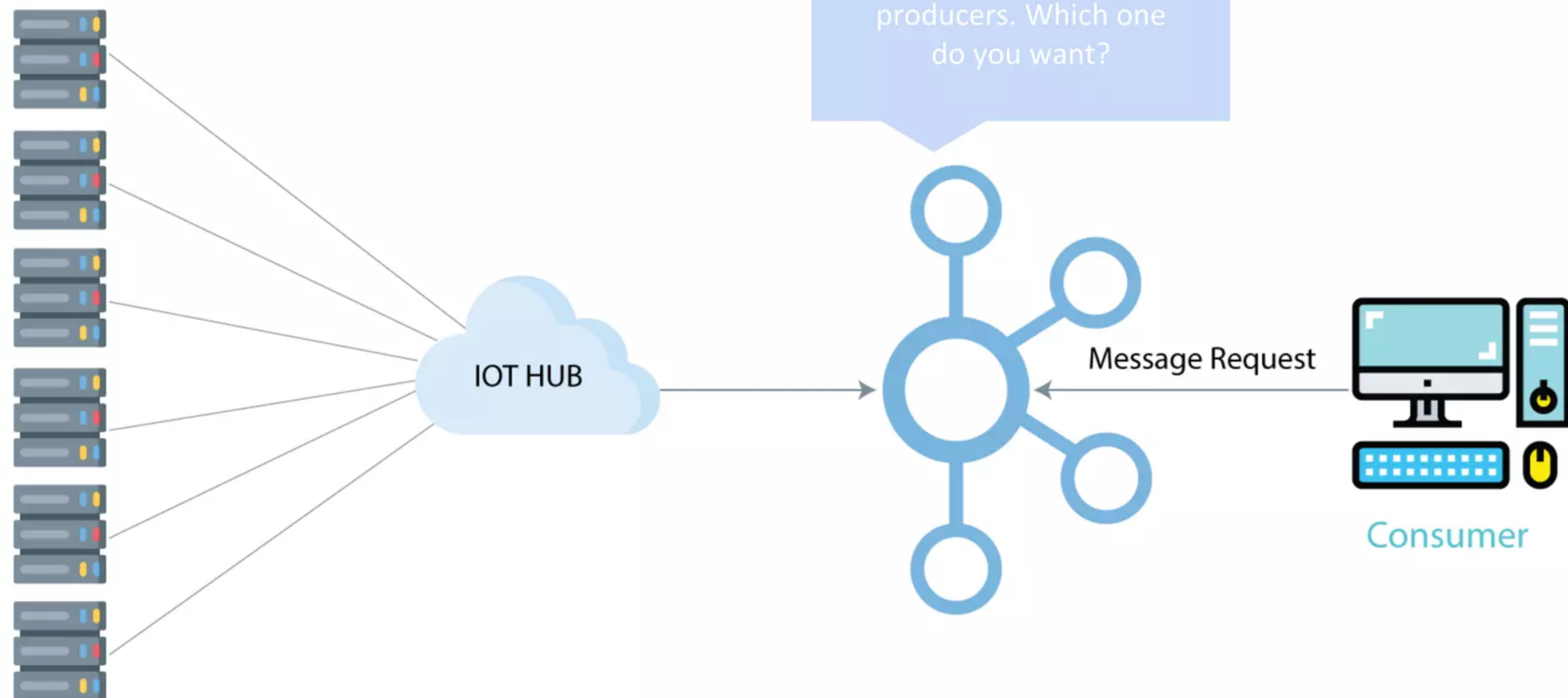
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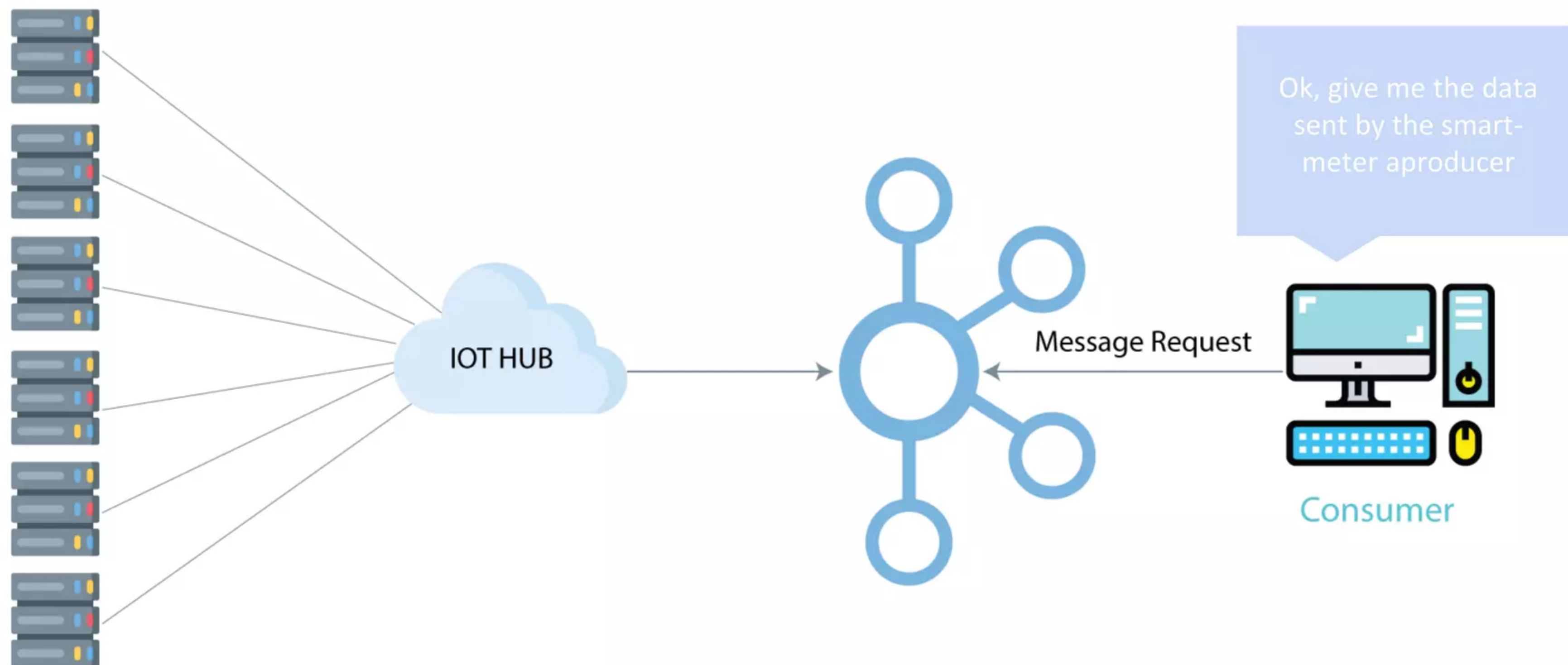
Agenda

- What is Apache Kafka?
- How Kafka works?
- History
- Kafka Products
- Kafka Core Components

Kafka Topic



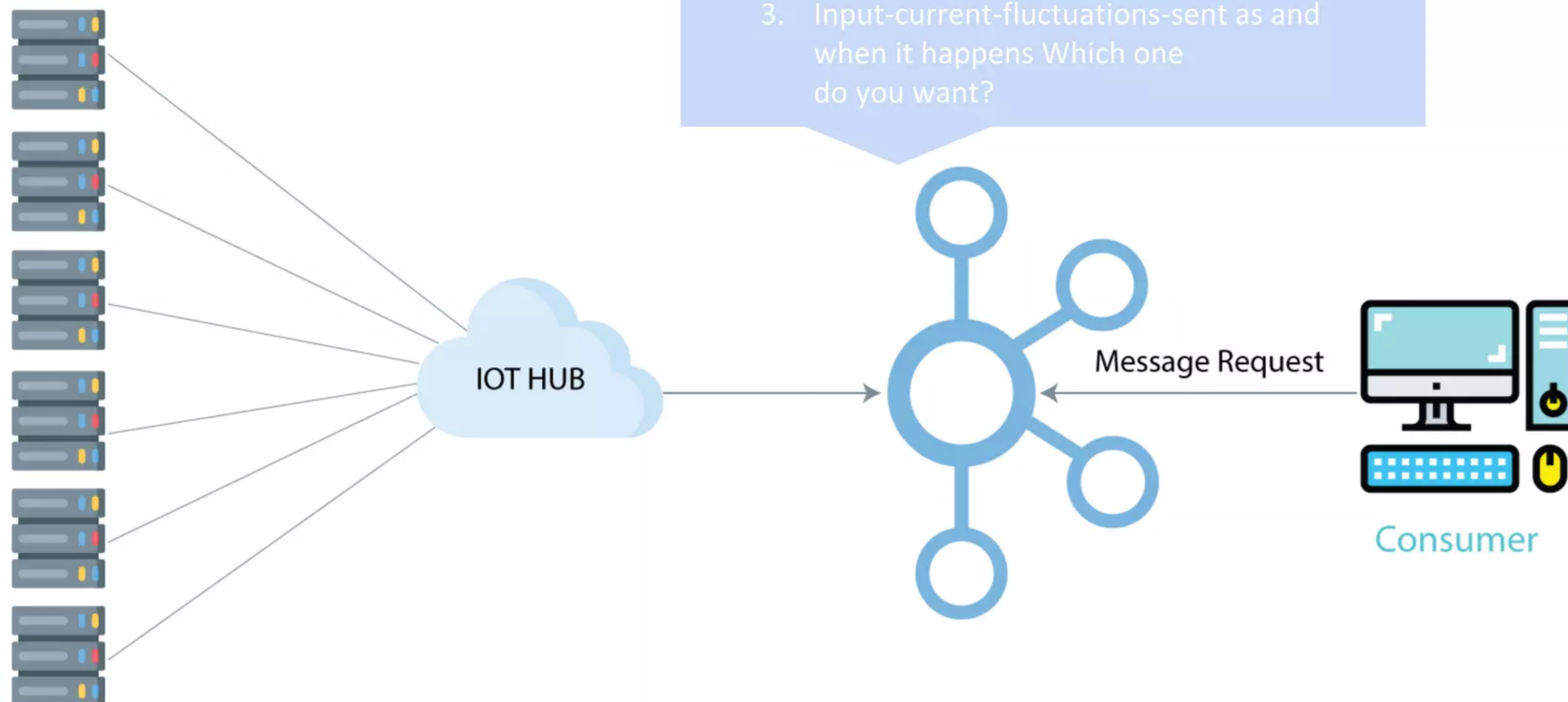
Kafka Topic



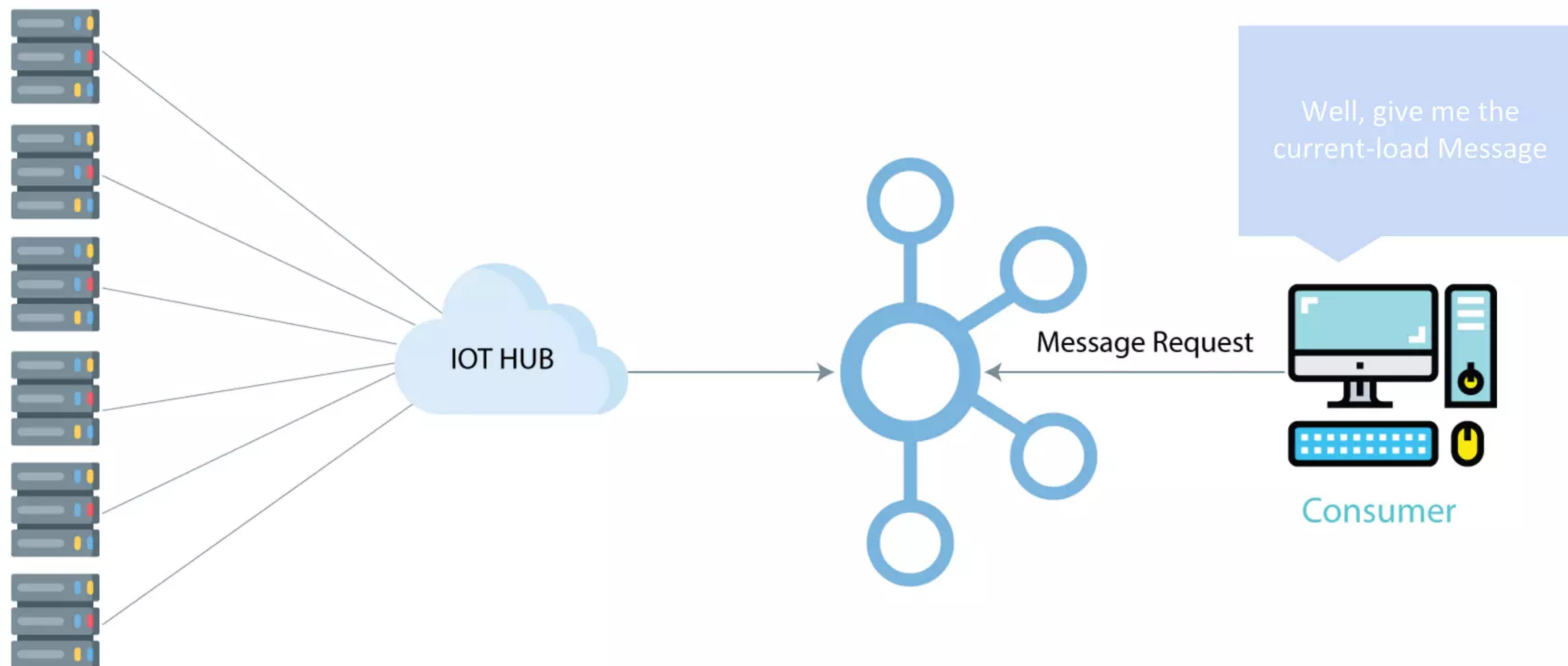
Kafka Topic

Well, the smart meter producer is sending three different types of data.

1. Current-load-sent every minute
2. Consumed-units-sent every hour
3. Input-current-fluctuations-sent as and when it happens Which one do you want?

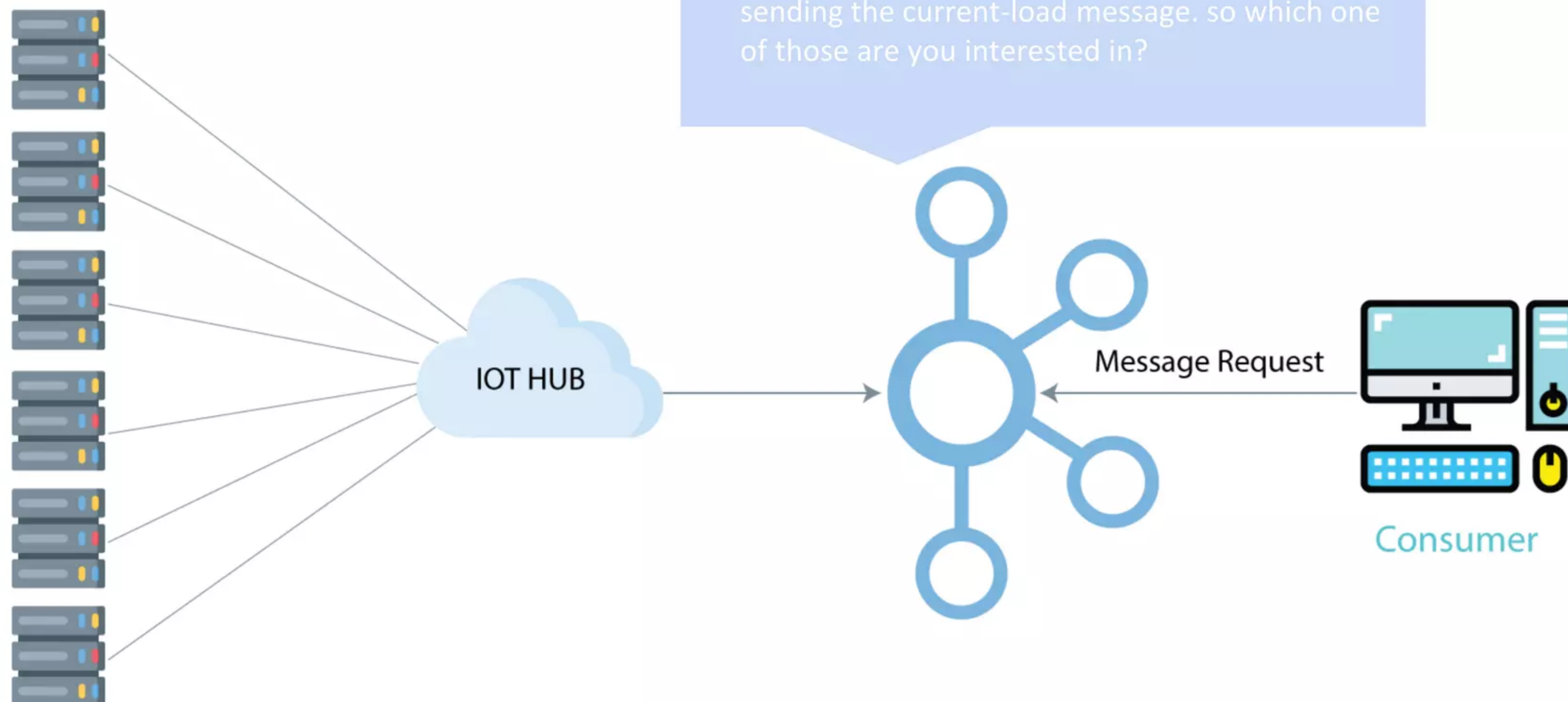


Kafka Topic

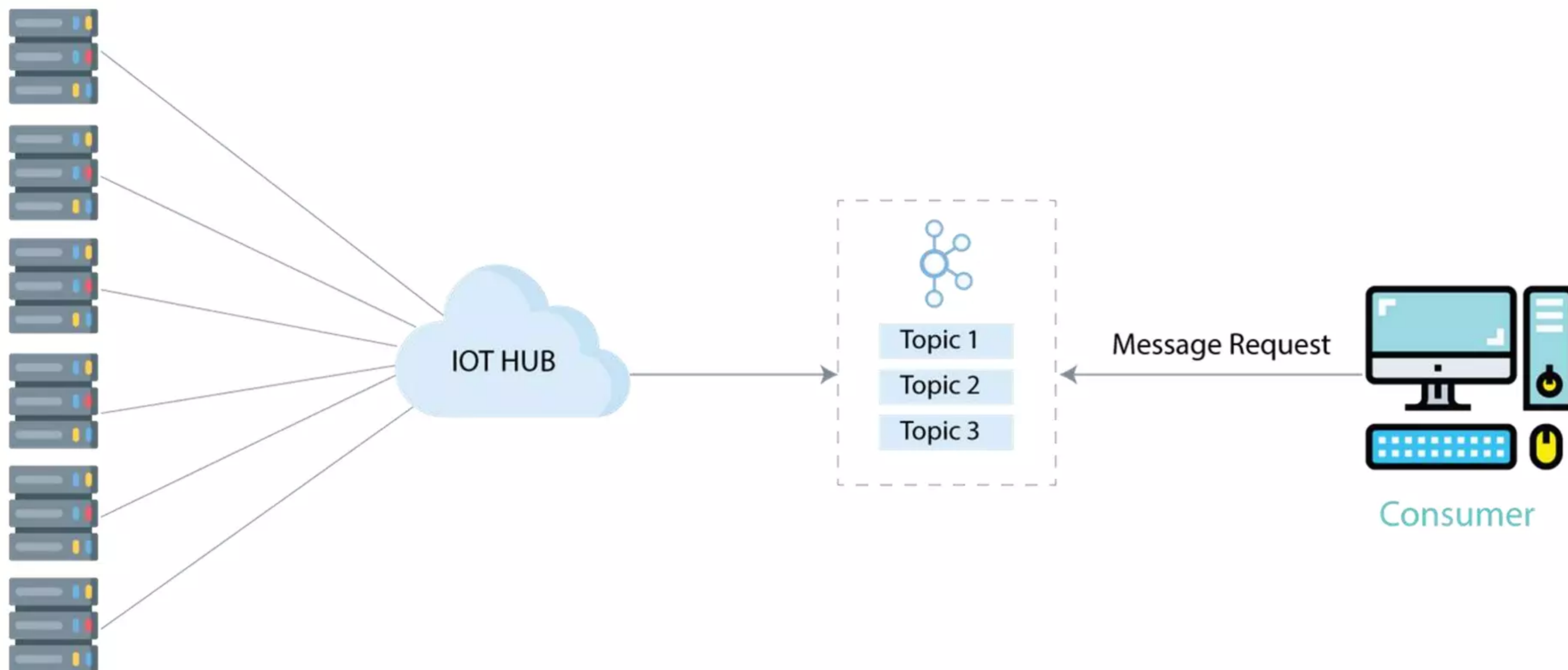


Kafka Topic

Ok, so you are looking for a current-load message. That's great! But we have a bunch of smart meter producers, and all of them are sending the current-load message. so which one of those are you interested in?



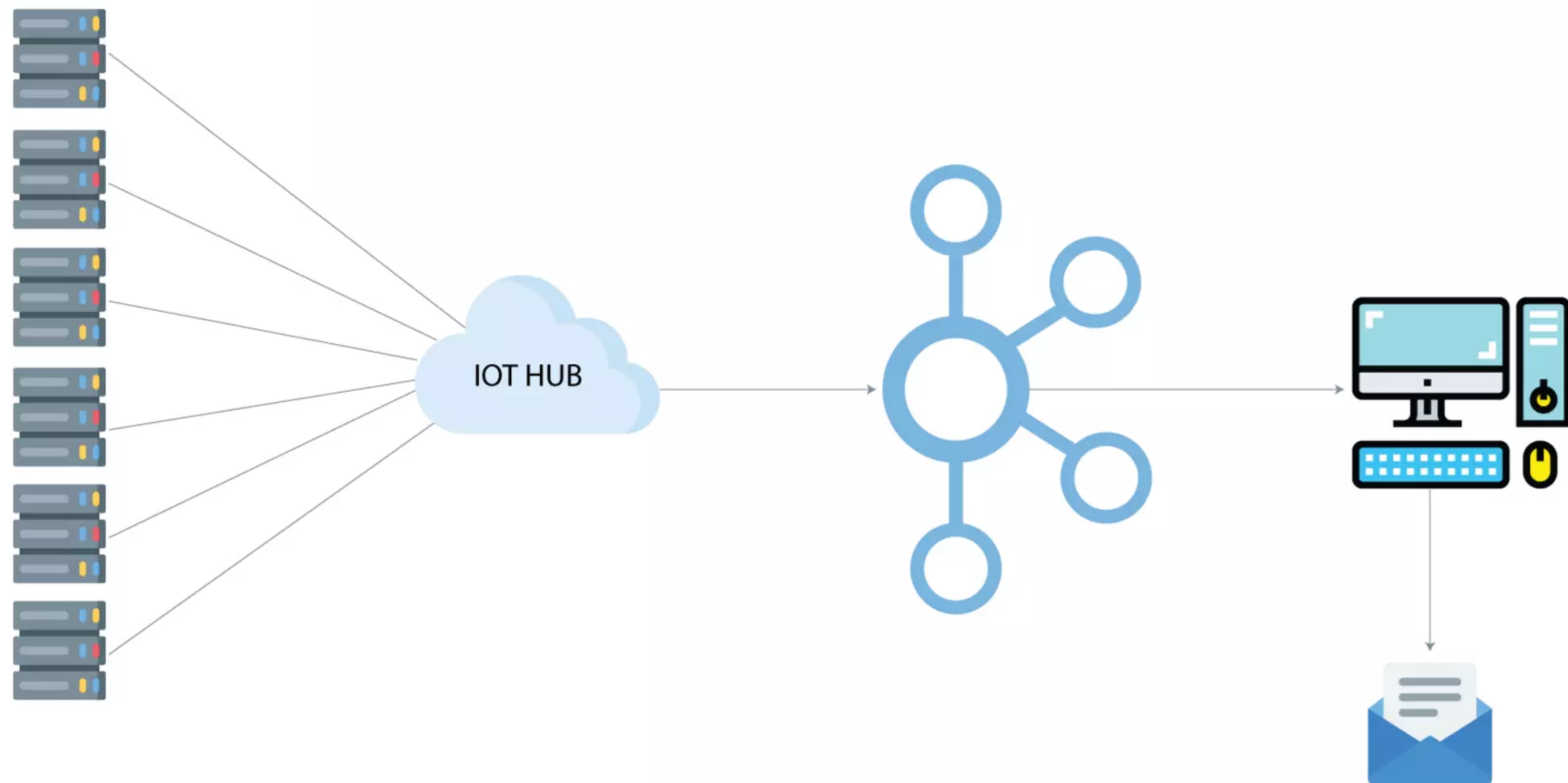
Kafka Topic



What is Apache Kafka?

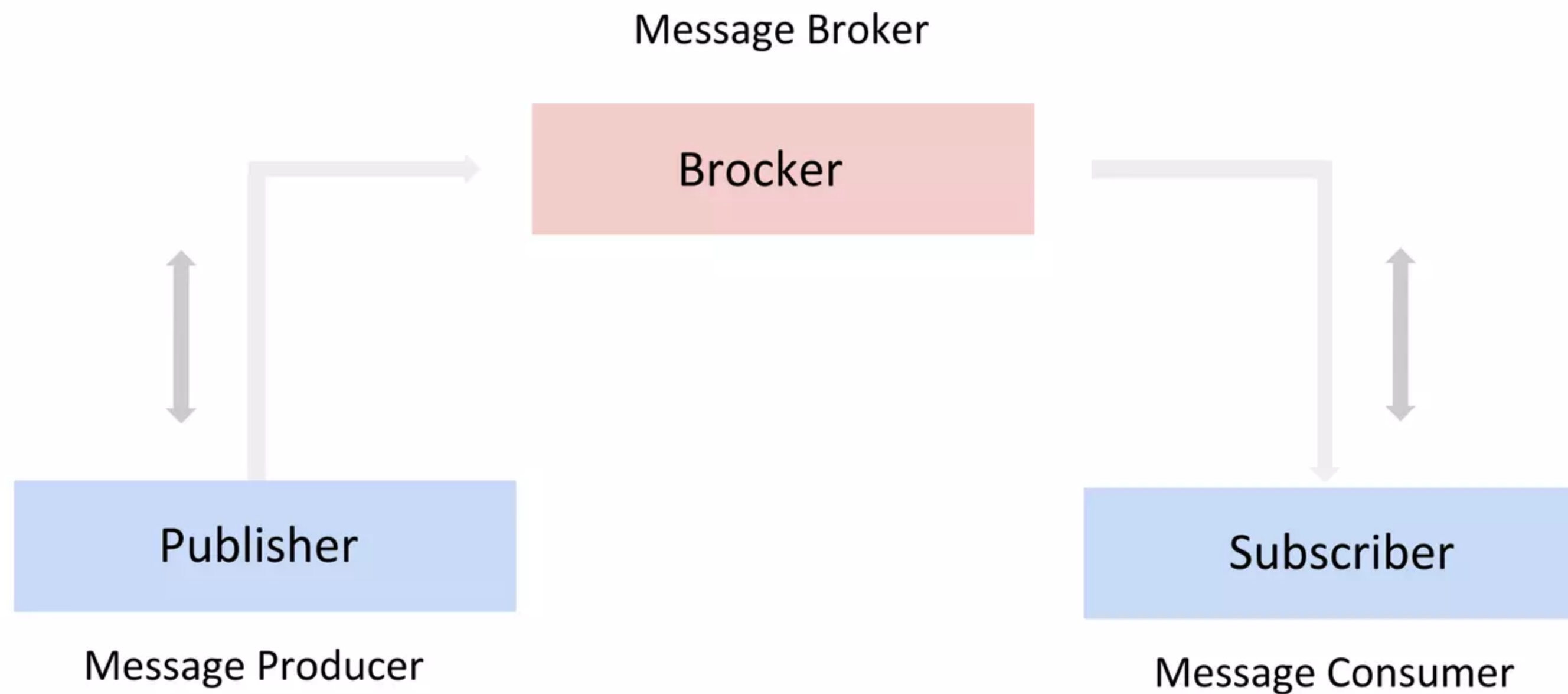
- Apache Kafka is a **distributed streaming** Open source platform developed by Apache Software Foundation, written in Scala and Java
 - A distributed system is a system with multiple components located on different machines that communicate and coordinate actions via messages in order to appear as a single system to the end-user.
 - Create real time data streams
 - Processing real time data stream

What is Apache Kafka?



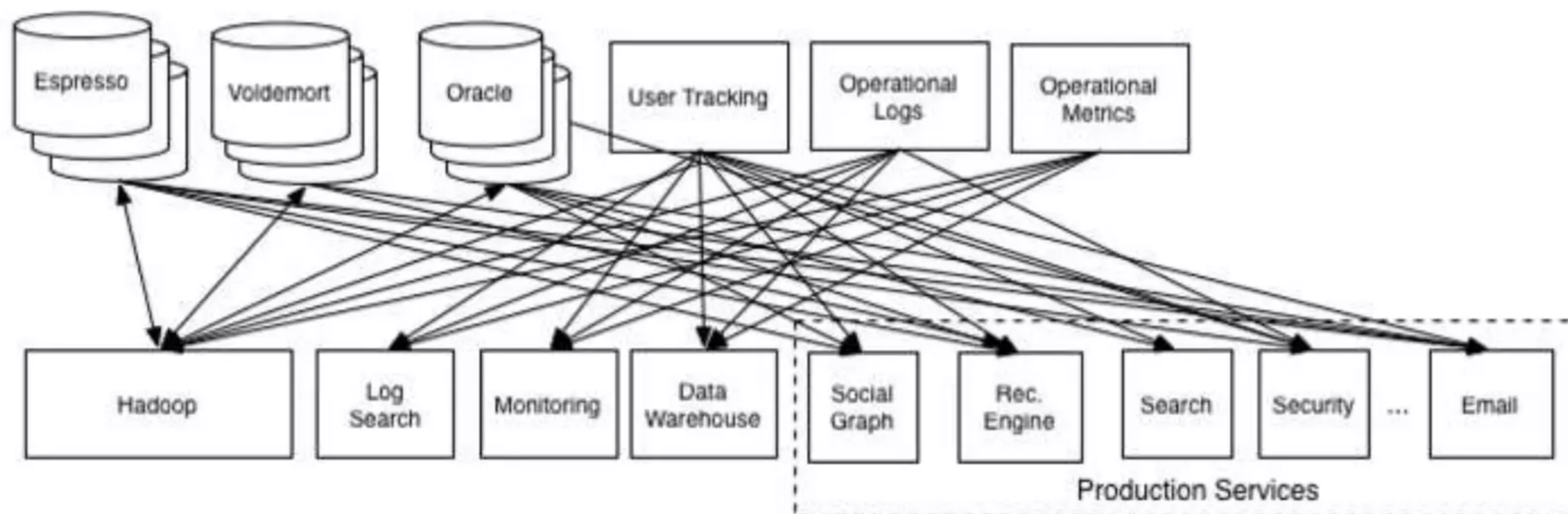
How Kafka Works?

- Kafka adopted Pub-Sub messaging architecture.

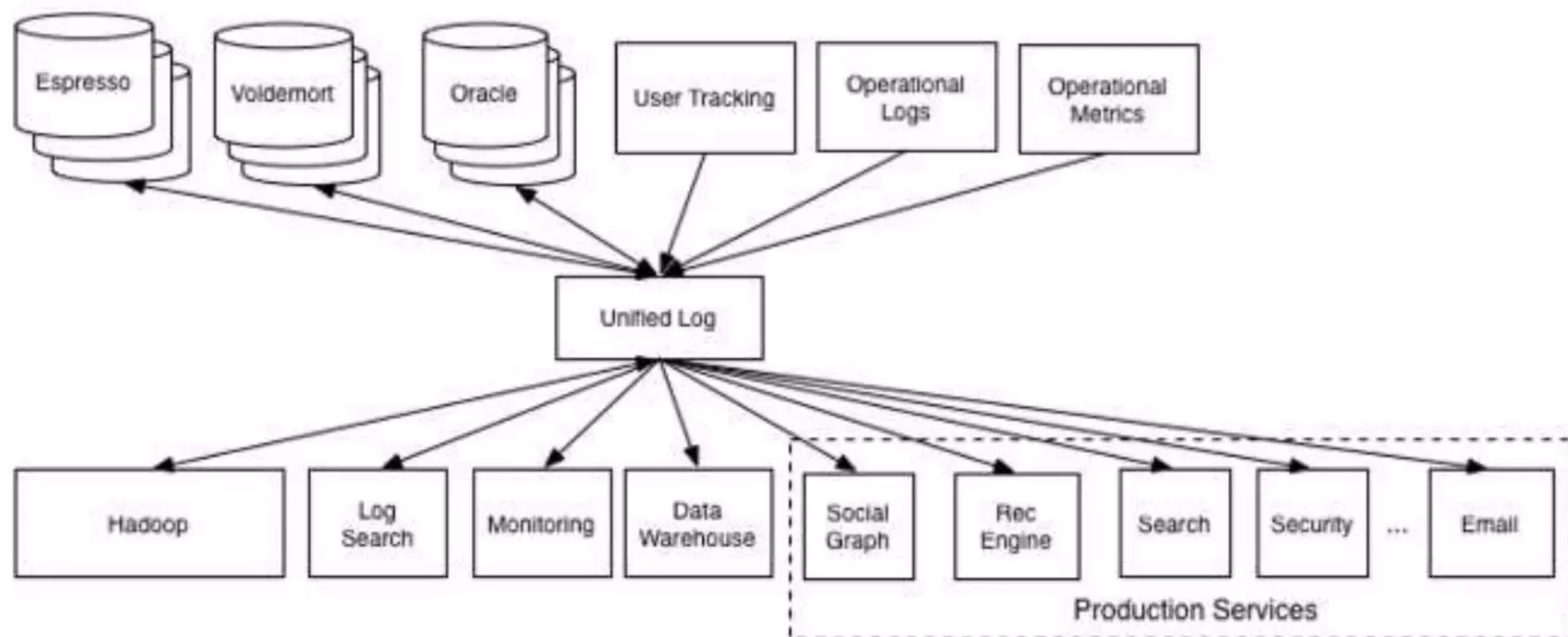


History

- Developed by LinkedIn in 2011 to solve data integration problem



History



Kafka Products

- **Kafka Broker** - Central server system
- **Kafka Client** - Producer & Consumer Java & Scala Library
- **Kafka Connect** - Data Integration tool
- **Kafka Stream** - Java & Scala library for Real time stream processing application
- **KSQL**

Kafka Core Concept

- Producer
- Consumer
- Broker/Server
- Cluster
- Topic
- Partition
- Offset
- Consumer Groups

Kafka Producer, Consumer, Broker

